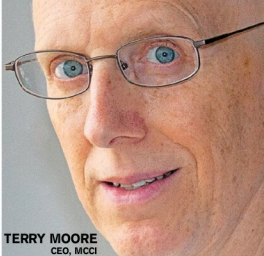


Embedded forms of USB will be very important for connecting modem chips to baseband processors

Some time ago, Dilin Anand from EFY spoke to Terry Moore, CEO, MCCI, a system engineering company which specialises in delivering USB system software to tier-1 silicon or semiconductor firms. MCCI chaired the committee that set up Mobile Broadband Interface Model (MBIM) specification and they also helped set up NCM specification, besides being principle author of the WMC family specifications

TERRY MOORE
CEO, MCCI



Where is USB positioned in the embedded wireless communications space?

The modem is almost all there's going to be as a separate module from the application processor. For most of the Internet of Things (IoT) applications, like the smart power meter and personal medical equipment, the volumes of individual products will be lower, and they want to ship international and they don't want to have to redesign and recertify it for every different market. Thus, they'll design a modem with a separate module where there will be communication through different variants of USB.

How should engineers select a solution for their USB implementation?

Typically they should start by selecting the required device controller or host controller driver from our library and then use a framework to produce a full-on USB solution. Next, on top of this framework they can add device-class protocols and class drivers to do the particular function that they want.

What is the most exciting aspect of USB Technology?

The exciting thing right now in consumer market is Media Agnostic USB (MA-USB), and in the embedded market it is SSIC-USB.

What are the major differences between MA-USB and the old wireless USB?

First of all it's based on Wi-Fi, the radio here is a Wi-Fi radio and is not something that's specially developed for USB. It is very fast and there is no or much less cell-to-cell interference.

The second important thing is that, instead of trying to model it on a packet-by-packet basis, they do it at the transfer level. In USB 2.0 and USB 3.0, each piece gets individually acknowledged; with MA-USB the acknowledgements are done at the same level and not at the level of the host controller. This makes it much more efficient.

Also instead of using their own pairing technique, they just leverage existing technology like Wi-Fi's P2P. Finally, they also leverage much cleaner Wi-Fi Max power management.

What would be the advantage of selecting this technology in consumer electronics design?

This is possibly a superior connectivity technology to something like Bluetooth or Wi-Fi Direct. It will work on any device, as long as the lower MAC has the right capability. With this software module, you can either appear as a device, or as a USB host, or even both.

What about the inter-chip solutions available with USB?

For LTE modems, the speeds are currently 150mbit/s. The USB Implementers Forum (USB-IF) has invented the Super-speed inter-chip USB, which is a chip-to-chip connection. One of the things that they developed was a very-low-power physical layer (PHY), called the M-PHY, which runs at speeds from 1 gigabit/s to 5 gigabit/s. However, SSIC-USB is ahead because, a lot of deployment of USB in mobile devices has already happened while there is no deployment of the others. But the speeds are naturally different since this is at an IC level. While Super-speed can achieve 5 gigabit/s, SSIC-USB is variable between 1 gigabit/s and 4 gigabit/s.

How is HSIC-USB different from conventional USB?

HSIC-USB is a specialised version of high-speed USB, optimised for low-power chip-to-chip applications. Conventional USB 2.0 devices use low-level signalling that works for 5-metre cables but is very power hungry; HSIC-USB uses low-level signalling that is extremely low-power, but only operates over 10cm distances on a PC. As a side effect, HSIC-USB is substantially more difficult to measure than normal USB.

Where will the most significant growth occur in emerging technologies?

There are two very important trends: the rise of the tablet, and the deployment of IoT. With respect to connectivity for tablet and mobile computing, the big area for growth right now is in 'wireless tethering' technologies. In the area of IoT, it's clear that embedded forms of USB will be very important for connecting modem chips to baseband processors. ●